Cloud Architecture

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GCXONE is built on modern cloud-native principles to provide a unified Video Surveillance as a Service (VSaaS) and IoT security management ecosystem.

1. The Core VSaaS Model

GCXONE operates entirely in the cloud, eliminating the complexity of traditional on-premise systems. By shifting to a SaaS model, organizations gain agility, scalability, and freedom from hardware maintenance.

No Local Servers: Eliminates dedicated local NVR/DVR maintenance.

Infinite Scalability: Add sites and cameras instantly without hitting physical hardware limits.

Centralized Management: Deploy security patches and features globally in seconds.

2. Infrastructure Stack (AWS)

The platform is hosted on Amazon Web Services (AWS) using a containerized microservices architecture managed by Kubernetes (EKS).

Compute & Orchestration

Private EKS Clusters: Core application logic runs on secure, private worker nodes.

Auto-Scaling: Node groups adjust resources in real-time based on traffic and processing load.

Messaging: Uses Amazon MQ for asynchronous event synchronization across the global platform.

Storage & Data

Asset Storage: Snapshots and icons are stored in encrypted S3 Buckets.

Metadata Management: Uses scalable NoSQL databases for lightning-fast camera metadata retrieval.

3. The Universal Translator (Proxy Layer)

The Proxy Architecture is GCXONE's core innovation. It allows the platform to communicate with diverse hardware manufacturers — Hikvision, Dahua, Axis, and others — by translating their proprietary protocols into a standardized unified command set.

The flow works as follows: Diverse hardware communicates via SDK or ONVIF to the Proxy Layer, which acts as a translation engine. The Proxy Layer converts this into a standardized stream using WebRTC or RTSP, which then feeds into the GCXONE Core for AI processing and display.

The Proxy Layer removes the need for expensive hardware bridges. Your cameras talk directly to the cloud via the GCXONE gateway.

4. Multi-Tenant Logical Hierarchy

Data isolation is enforced through a strict multi-tenant model. Each organization lives in a dedicated Tenant space, ensuring data privacy and granular access control.

The hierarchy is structured as follows:

Tenant: The monitoring station or master organization.

Customer: Individual client organizations under the tenant.

Site: Physical facilities such as offices or warehouses.

Device: NVRs or Gateways located at the site.

Sensor: Individual camera channels or PIR sensors connected to a device.

5. Security & Compliance

Security is integrated at the architectural level, not added as a bolt-on feature.

Encryption: Data at rest and in transit is secured with AES-256.

Access Gating: Jump Servers and private VPCs isolate infrastructure from the public web.

Authentication: Integrated with Auth0 for enterprise-grade MFA and SSO.

Continuous Audit: Every system call and operator action is logged in an immutable audit trail.